

Python 3.13.13 (tags/v3.13.13:01104ce, Apr 7 2026, 19:25:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/io.py =====
Water Jug Solution

Initial State -> (0,0)
Fill 3-gallon Jug -> (0,3)
Pour 3-gallon into 4-gallon -> (3,0)
Fill 3-gallon Jug Again -> (3,3)
Pour into 4-gallon until full -> (4,2)
Empty 4-gallon Jug -> (0,2)
Pour remaining water into 4-gallon -> (2,0)

Goal Achieved!
Exactly 2 gallons are in the 4-gallon jug.

===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/io.py =====
Mars Rover Mission Started

Moving to a new location...
Capturing surface image...
Collecting soil sample...
Analyzing collected sample...
Sending data to Earth...

Mission Summary

Location : Landing Site
Samples Collected : 1

===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/io.py =====
Solution Found:

[1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0]
[0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1]
[0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0]

>>>

===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/io.py =====
Enter Pickup Location: KKD NAGAR
Enter Destination: VR MALL

Available Cab Types

1. Mini

2. Micro
3. Sedan
4. Shared
5. Prime

Select Cab Type: MICRO

Invalid Cab Selection

>>>

===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/io.py =====

Least Cost Path : S -> A -> C -> D -> G

Total Cost : 6